Introduction to ABAP Core Data Services

ABAP Platform Product Management, SAP SE
May 2024

PUBLIC

Agenda

MODERN ABAP DEVELOPMENT

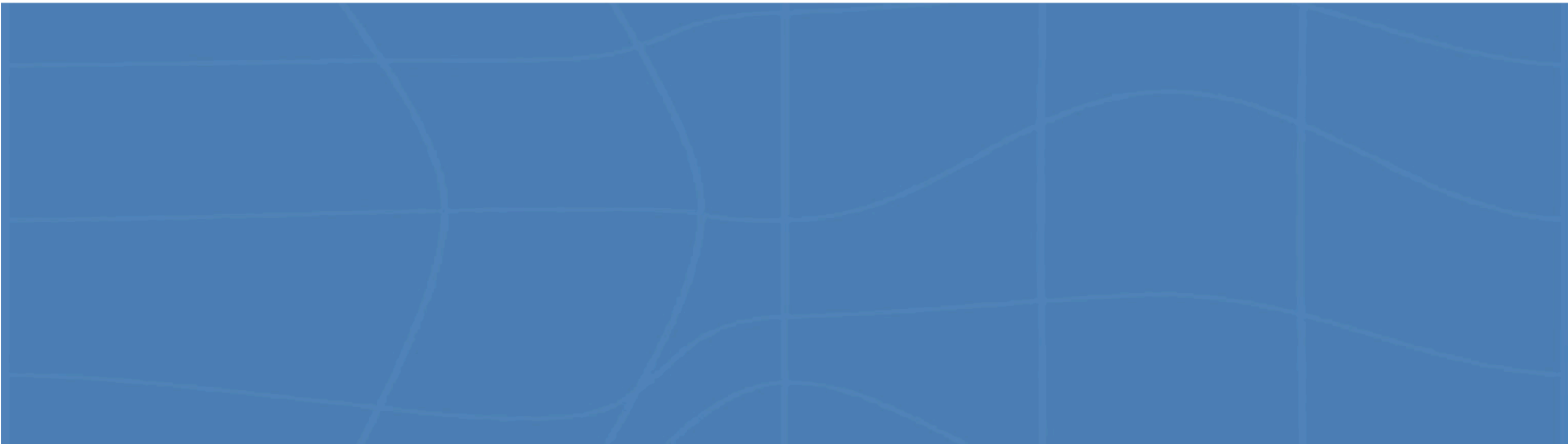
ABAP CORE DATA SERVICES (CDS) OVERVIEW

CDS IN ABAP APPLICATION PROGRAMMING

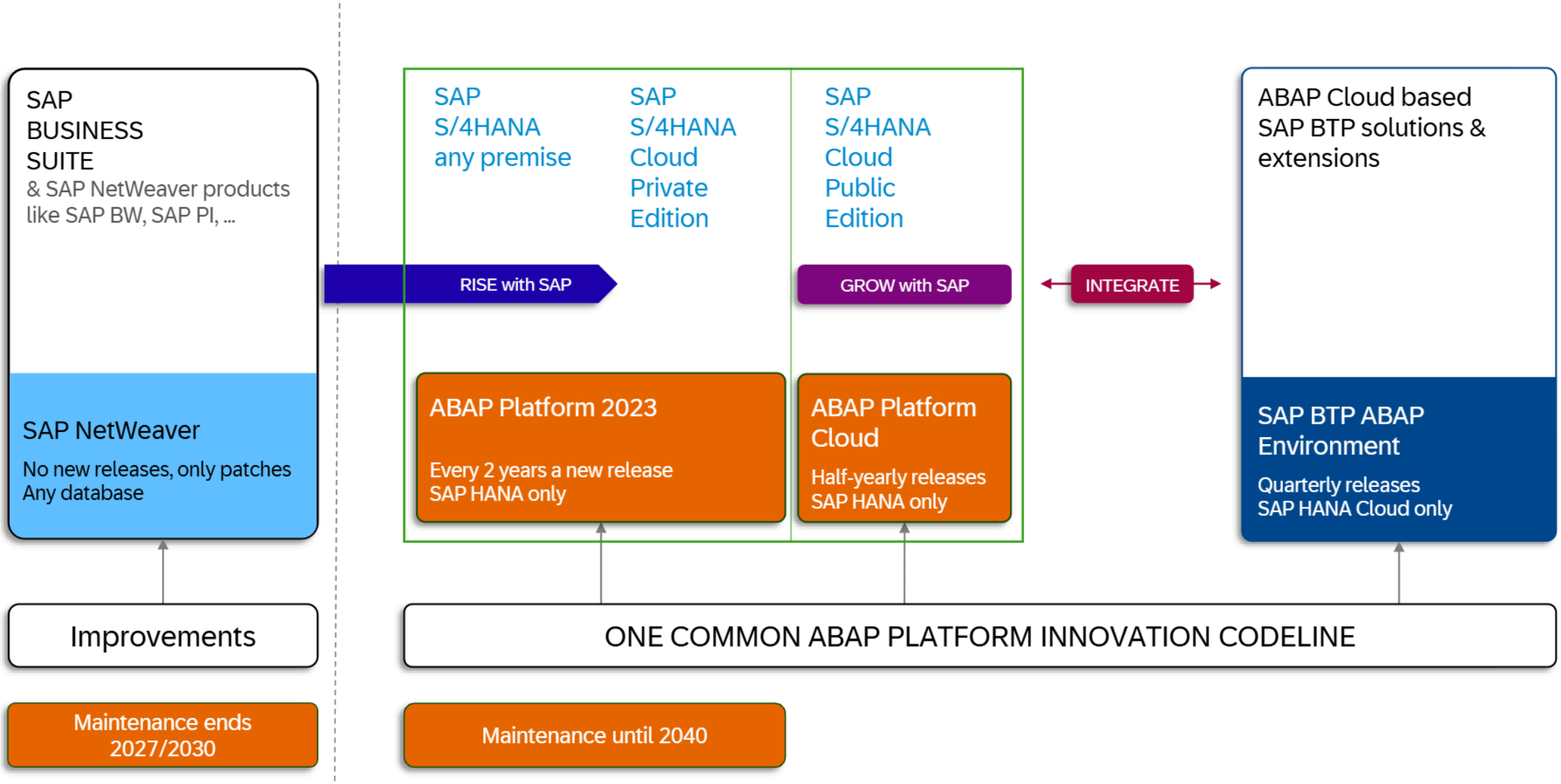
SUMMARY

PUBLIC

MODERN ABAP DEVELOPMENT



ABAP Platform – Foundation of SAP's ERP applications



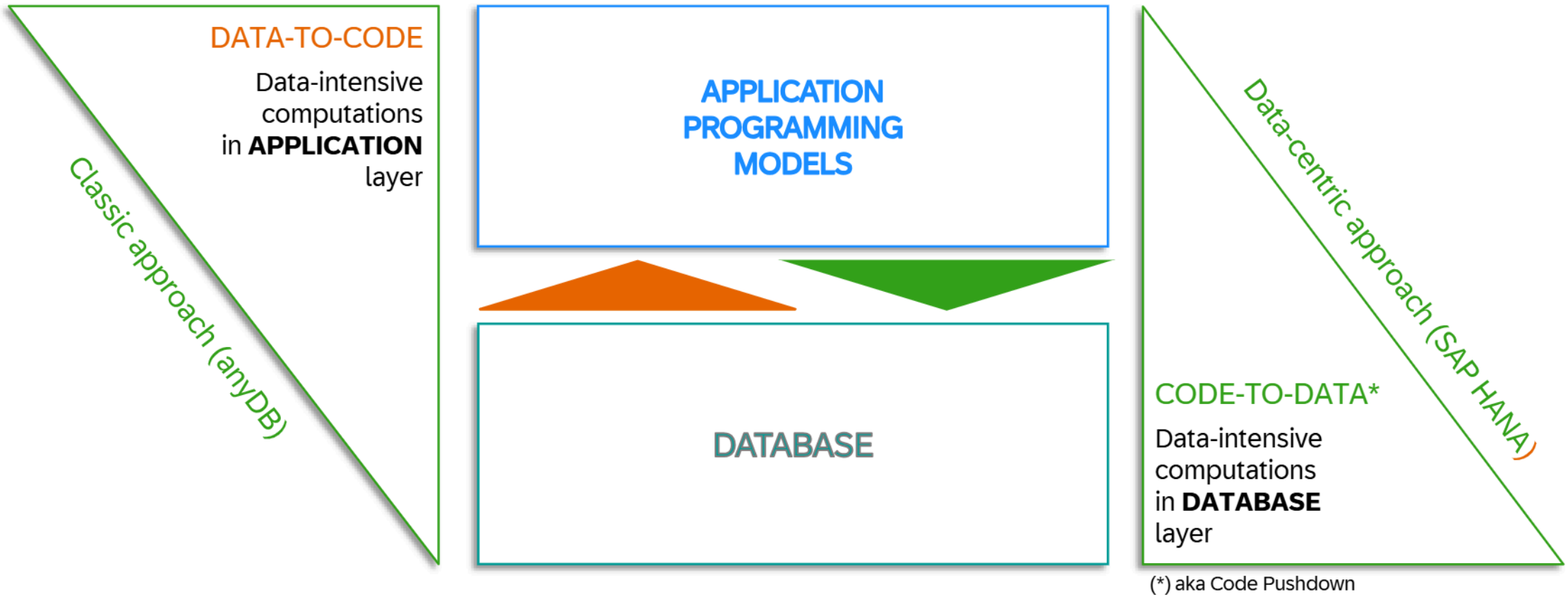
Modern ABAP application development

Must learn topics: UX, SAP HANA, state of the art development, Cloud

Focus on **User Experience**	Utilize **SAP HANA** capabilities	Efficient **development**	**Cloud**
On any device Role based Device switch and collaboration Browser based SAP Fiori and SAPUI5, ABAP RESTful Application Programming Model	ABAP-managed Code Pushdown Advanced application and Function libraries ABAP SQL, Core Data Services, SAP HANA breakouts	Secure app development, high code quality Agile & test driven Customer code management ABAP language and tools, Development methodology	Stable Cloud extensions In-app and side-by-side Released APIs Cloud ready apps In-App extensibility, Developer extensibility, SAP BTP

Modern ABAP application development

Programming paradigm shift powered by SAP HANA

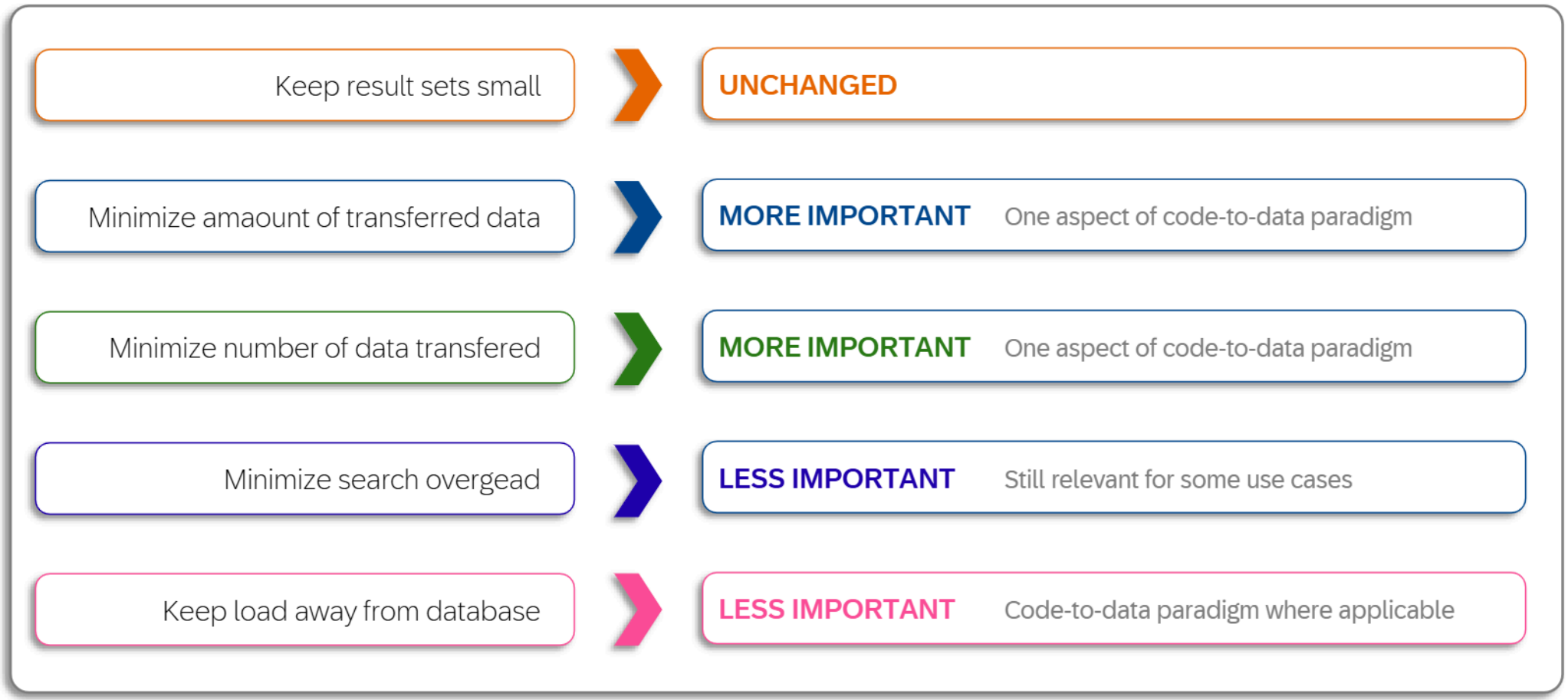


(*) aka Code Pushdown

More in this blog

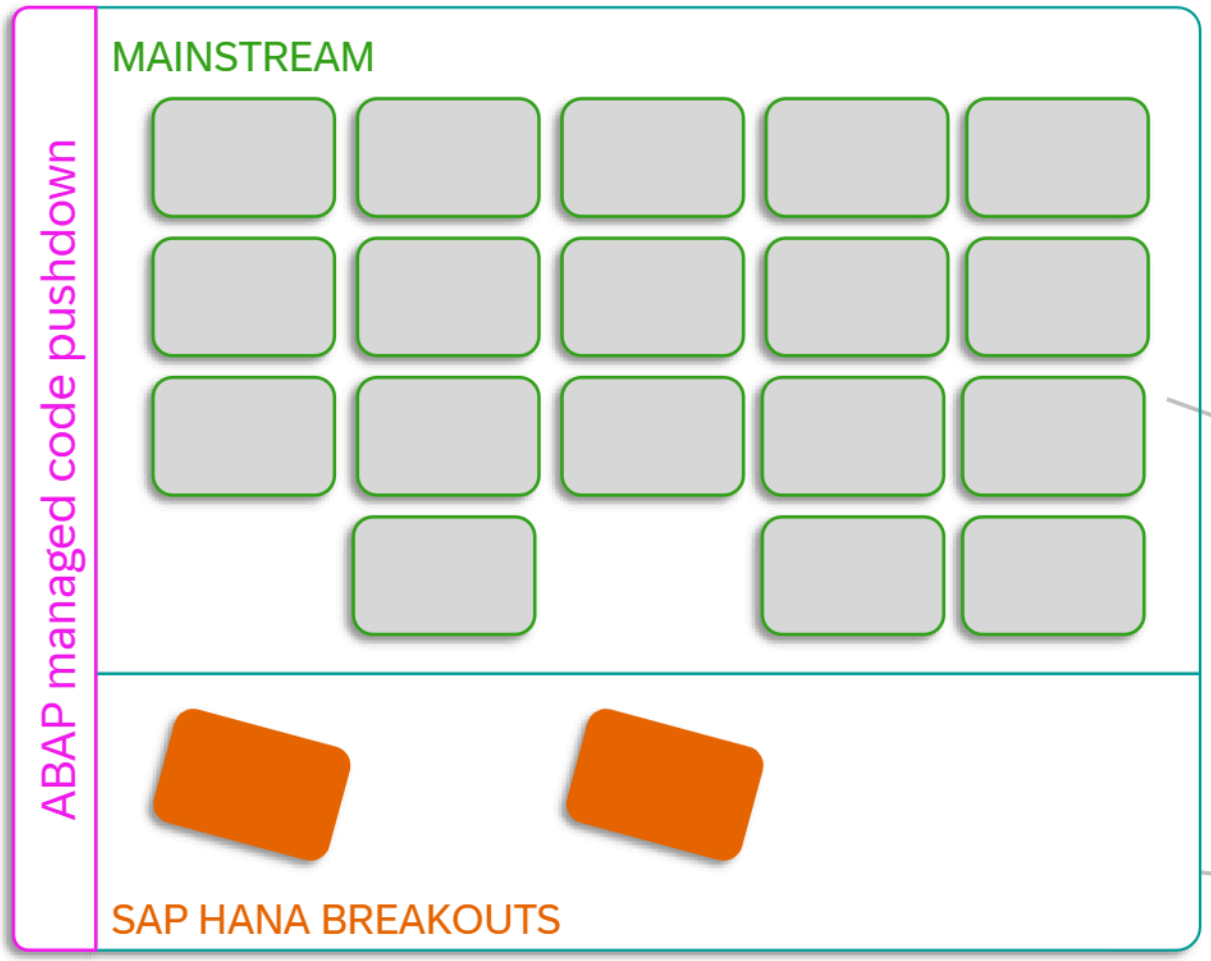
Modern ABAP application development

Golden rules for SQL programming – Priority shift on SAP HANA



More in this blog

Modern ABAP application development on SAP HANA



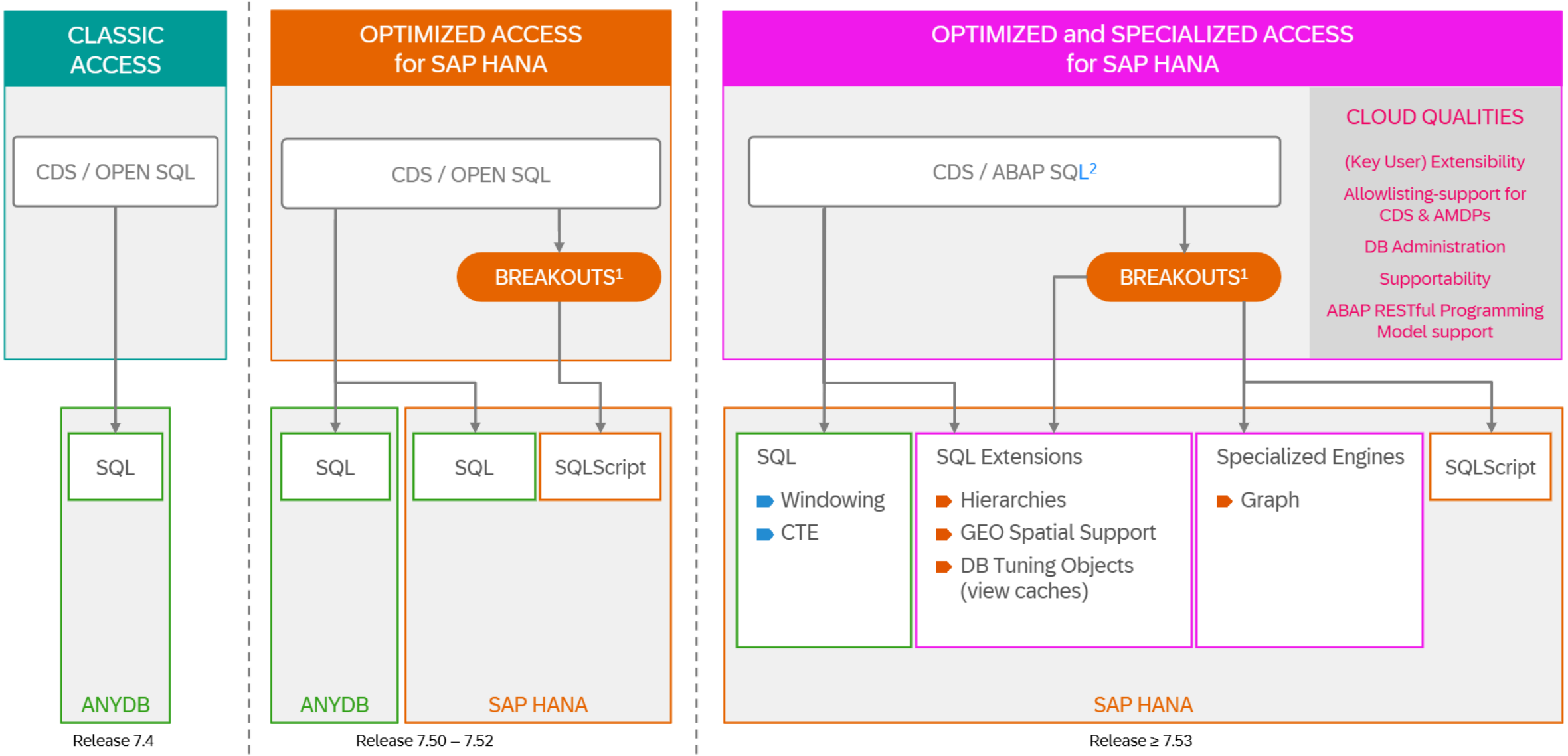
ABAP Core Data Services (CDS)

ABAP SQL (fka Open SQL)

ABAP Managed Database Procedures (AMDP) – Procedures and Functions

CDS Table Functions

Evolution of ABAP CDS and Open/ABAP SQL



[1] AMDP procedures and functions, CDS table functions, and CDS hierarchies

[2] Open SQL was renamed to ABAP SQL starting with ABAP Platform 1809

Efficient ABAP development in Eclipse

HIGH DEVELOPER PRODUCTIVITY ON-PREMISE AND IN THE CLOUD

MODERN DEVELOPMENT TOOLSET

Fully eclipse-based

Syntax check, code completion

Navigation, search, quick fixes

Full support for the ABAP RESTful Application Programming Model and CDS

QUALITY ASSURANCE

Static code checks (ATC, CVA) with remote and local scenarios

Unit testing incl. isolation frameworks

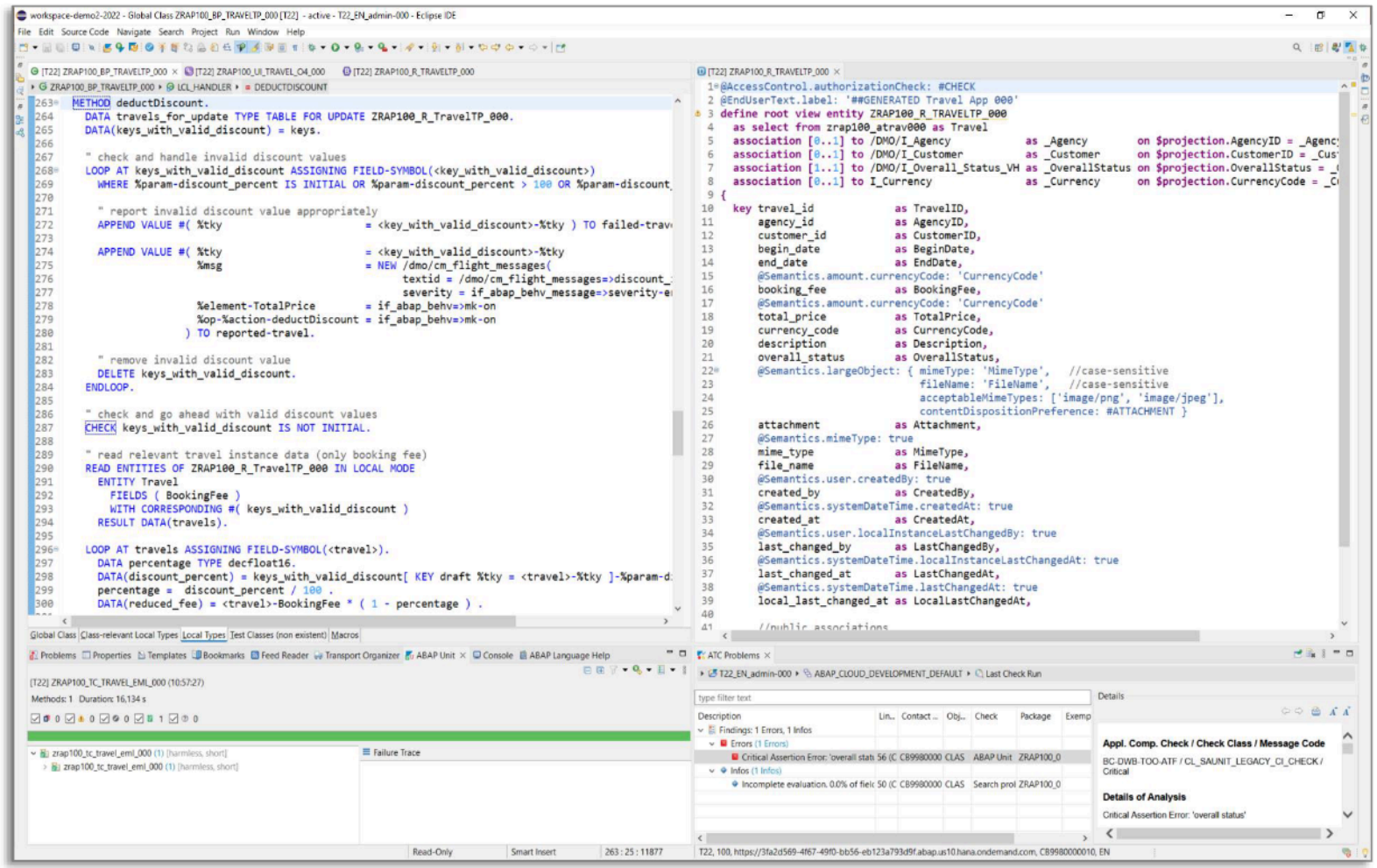
Test seams and injections

SUPPORTABILITY

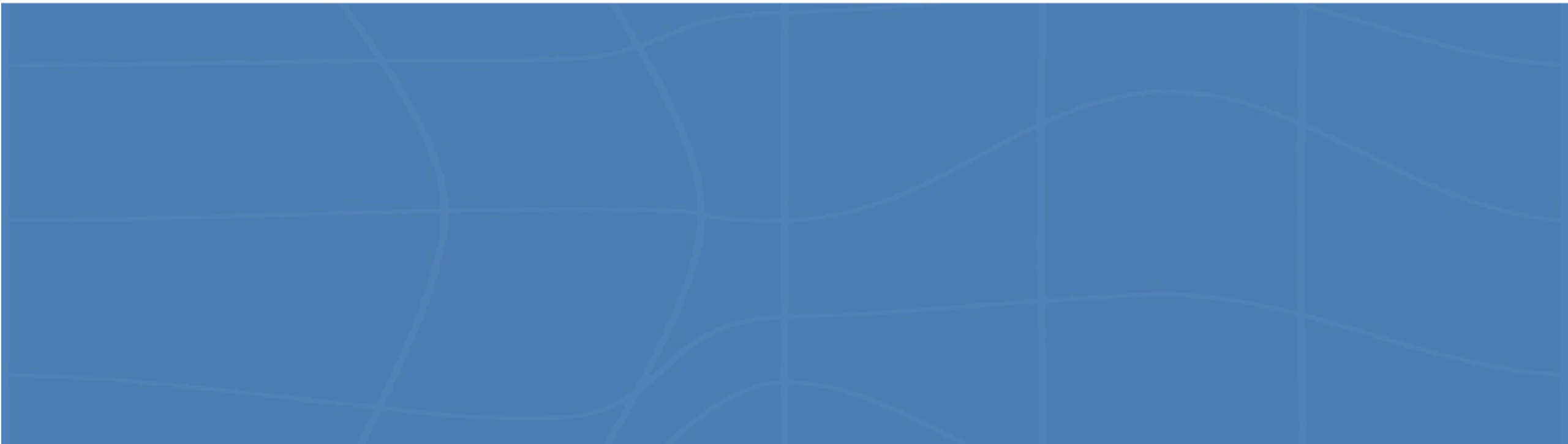
Debugging, profiling, tracing

Static and dynamic logging

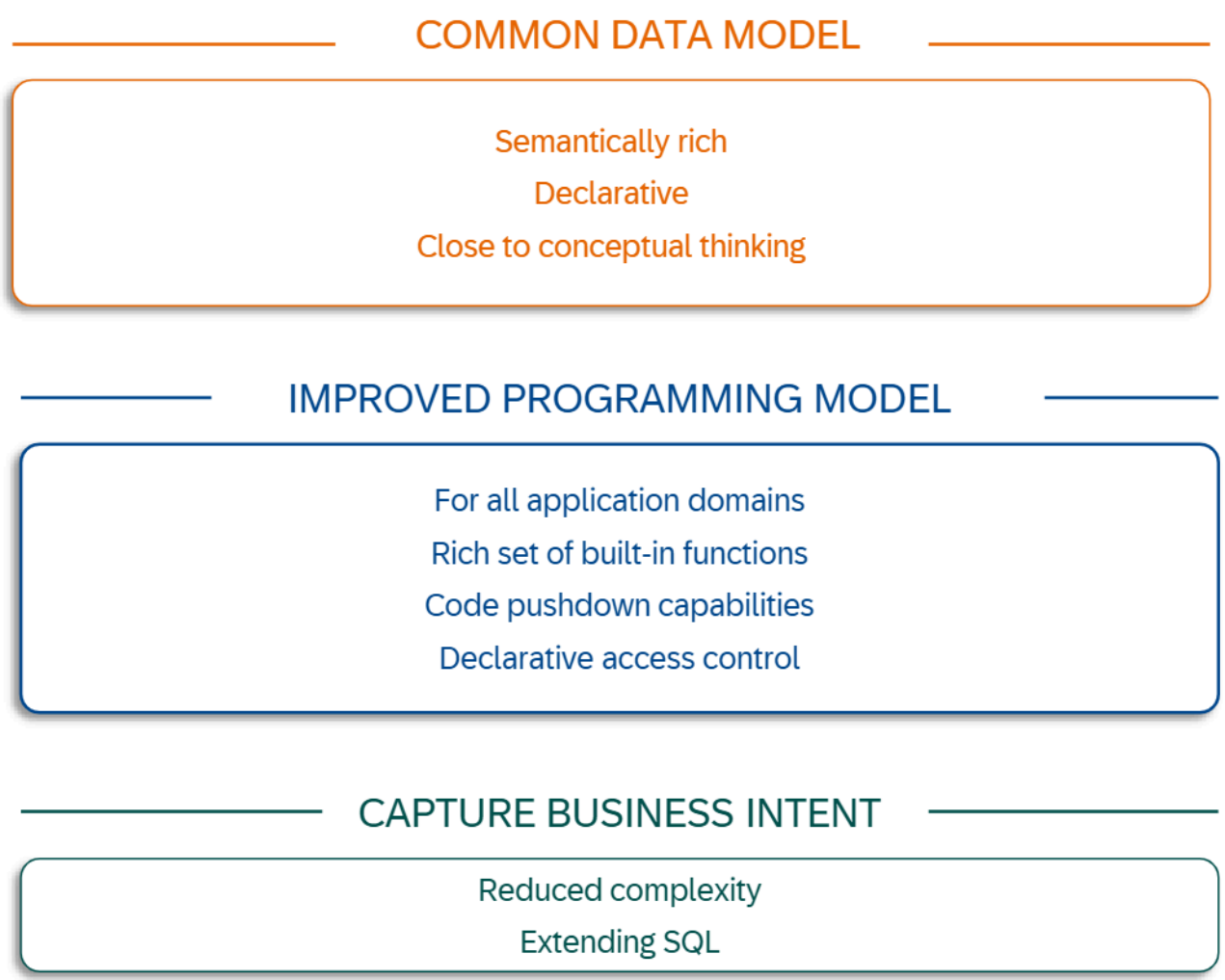
Runtime monitoring and analysis



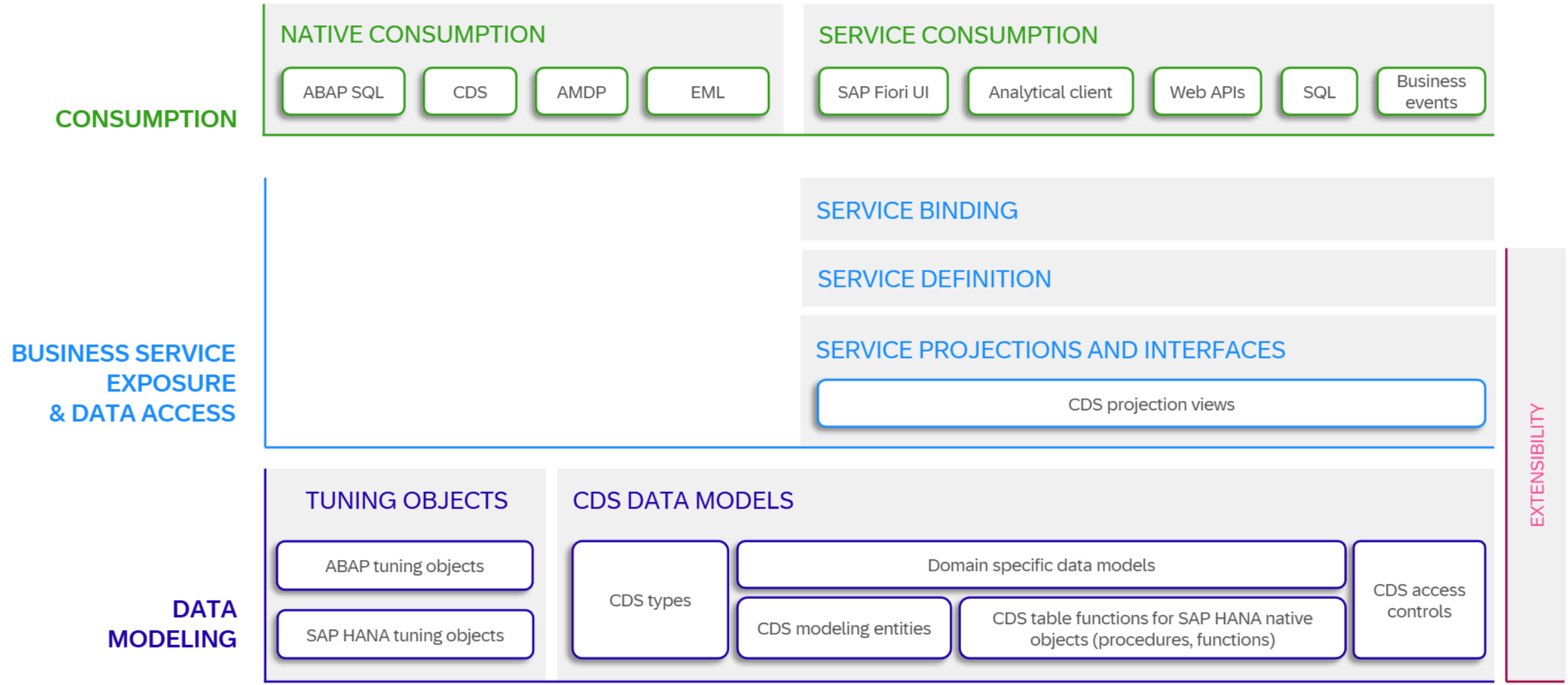
ABAP CDS OVERVIEW



ABAP CDS – Next generation data modeling and access



ABAP CDS data models



ABAP CDS – Overview of supported CDS entity types[1]

Category	ABAP CDS Entity Type	ABAP CDS Statement

Standard view building	CDS View Entity
	CDS Projection View
	CDS Projection View – Analytical Query
	CDS DDIC-based View *(deprecated)*
		`DEFINE VIEW ENTITY`
		`DEFINE VIEW ENTITY AS PROJECTION ON`
		`DEFINE TRANSIENT VIEW ENTITY AS PROJECTION ON`
		`DEFINE VIEW` *(deprecated)*

Advanced view building – External implementation	CDS Custom Entity CDS Abstract Entity	`DEFINE CUSTOM ENTITY` `DEFINE ABSTRACT ENTITY`

Advanced view building – SAP HANA breakout	CDS Table Function
	CDS Hierarchy
	CDS Scalar Function[2]
		`DEFINE TABLE FUNCTION`
		`DEFINE HIERARCHY`
		`DEFINE SCALAR FUNCTION`

Type definition	CDS Simple Type[2] CDS Enumerated Type[2]	`DEFINE SIMPLE TYPE` `DEFINE TYPE ENUM`
Transactional behavior definition of business objects built with the ABAP RESTful Application Programming Model	CDS Behavior Definition (and Projection)	`DEFINE BEHAVIOR`

Modification-free extension	CDS View Entity Extension
	CDS Custom Entity Extension
	CDS Abstract Entity Extension
	CDS DDIC-based View Extension *(deprecated)*
	CDS Metadata Extension
	CDS Behavior Extension
		`EXTEND VIEW ENTITY`
		`EXTEND CUSTOM ENTITY`
		`EXTEND ABSTRACT ENTITY`
		`EXTEND VIEW` *(deprecated)*
		`ANNOTATE VIEW`
		`EXTEND BEHAVIOR`

Access control definition	CDS Access Control	`DEFINE ROLE`

[1] Status from Q2/2024 | [2] Available on-prem starting with Release 2023

ABAP CDS view entity – Example

View annotations

View definition

Associations

Selection

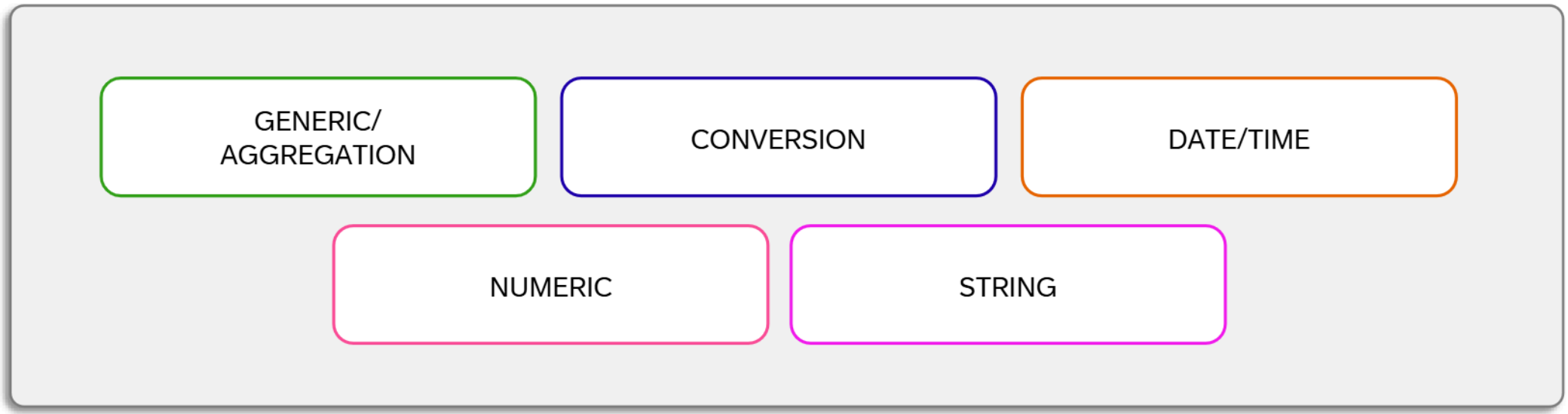
Data source

Element annotations

```
/DMO/I_TRAVEL_D
1  @EndUserText.label: 'Travel View Entity for Draft RefScen'
2  @AccessControl.authorizationCheck: #NOT_REQUIRED
3
4  define root view entity /DMO/I_Travel_D
5    as select from /dmo/a_travel_d
6
7    composition [0..*] of /DMO/I_Booking_D         as _Booking
8
9    association [0..1] to /DMO/I_Agency            as _Agency        on $projection.AgencyID = _Agency.AgencyID
10   association [0..1] to /DMO/I_Customer          as _Customer      on $projection.CustomerID = _Customer.CustomerID
11   association [1..1] to /DMO/I_Overall_Status_VH as _OverallStatus on $projection.OverallStatus = _OverallStatus.OverallStatus
12   association [0..1] to I_Currency               as _Currency      on $projection.CurrencyCode = _Currency.Currency
13
14 { ///dmo/a_travel_d
15   key travel_uuid           as TravelUUID,
16
17       travel_id             as TravelID,
18       agency_id             as AgencyID,
19       customer_id           as CustomerID,
20       begin_date            as BeginDate,
21       end_date              as EndDate,
22       @Semantics.amount.currencyCode: 'CurrencyCode'
23       booking_fee           as BookingFee,
24       @Semantics.amount.currencyCode: 'CurrencyCode'
25       total_price           as TotalPrice,
26       currency_code         as CurrencyCode,
27       description           as Description,
28       overall_status        as OverallStatus,
29
40       //local ETag field
41       @Semantics.systemDateTime.lastChangedAt: true
42       last_changed_at       as LastChangedAt,
43
44       //Associations
45       _Booking,
46
47       _Agency,
48       _Customer,
49       _OverallStatus,
50       _Currency
51 }
```

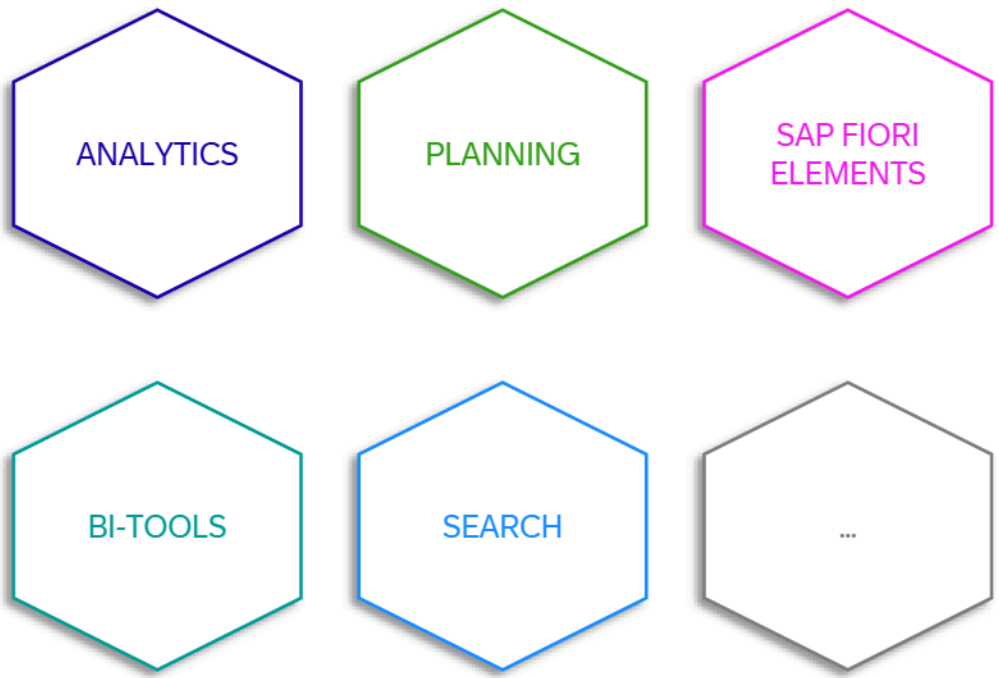
ABAP Flight Reference Scenario -
Example available in package /DMO/FLIGHT_DRAFT

Built-in SQL functions and expressions



CDS annotations for domain-specific frameworks

CDS ANNOTATION DOMAINS



Enrich meta model with consumer-specific vocabularies

Reusable and unified view model for all use cases

Enabling flexible usage in different contexts

Separation of concern using CDS metadata extensions for UI-specific annotations

Vital part of new ABAP programming models

CDS associations

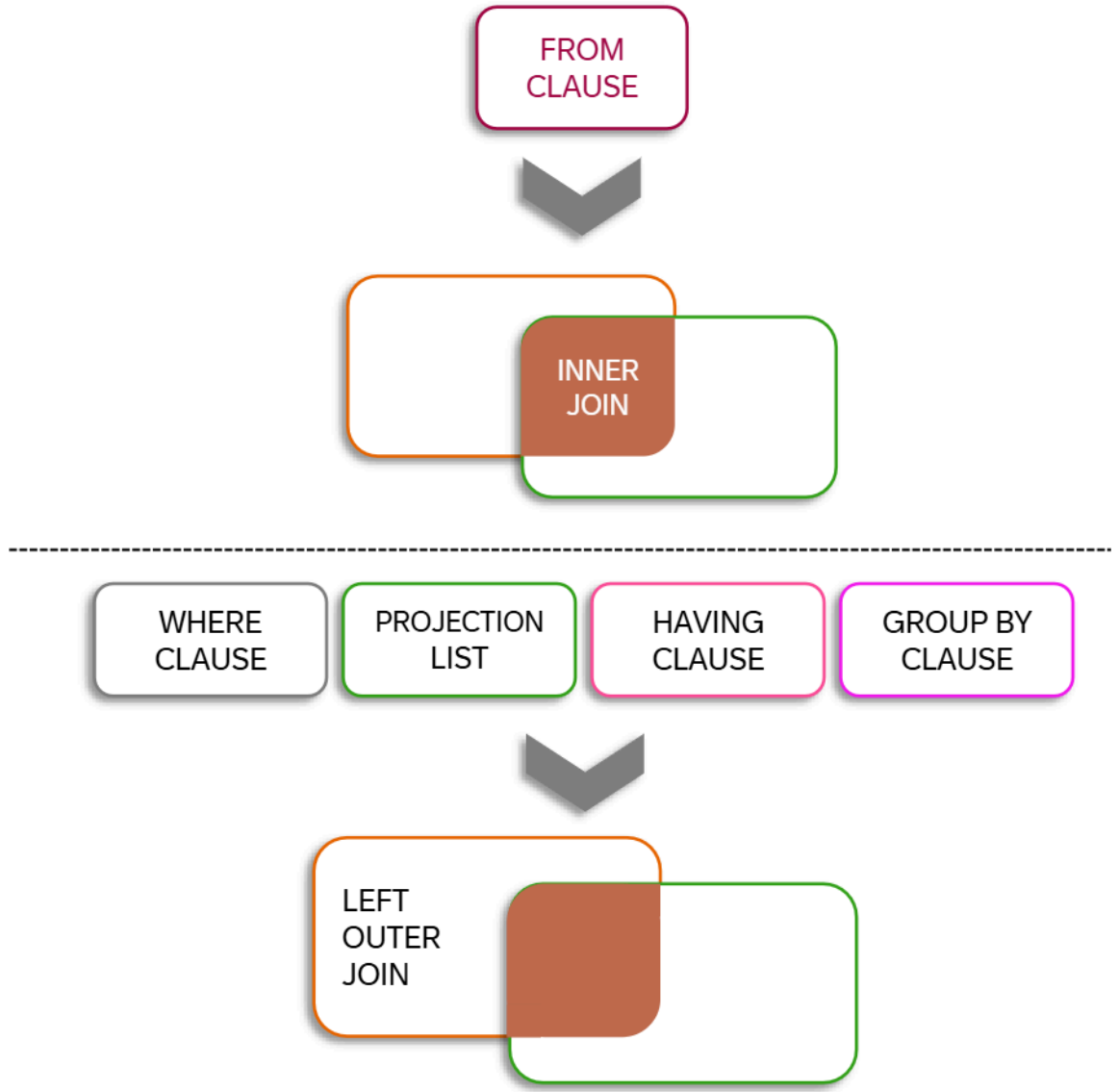
Capturing the relationships between entities in the data model

Used associations are implicitly translated into SQL joins

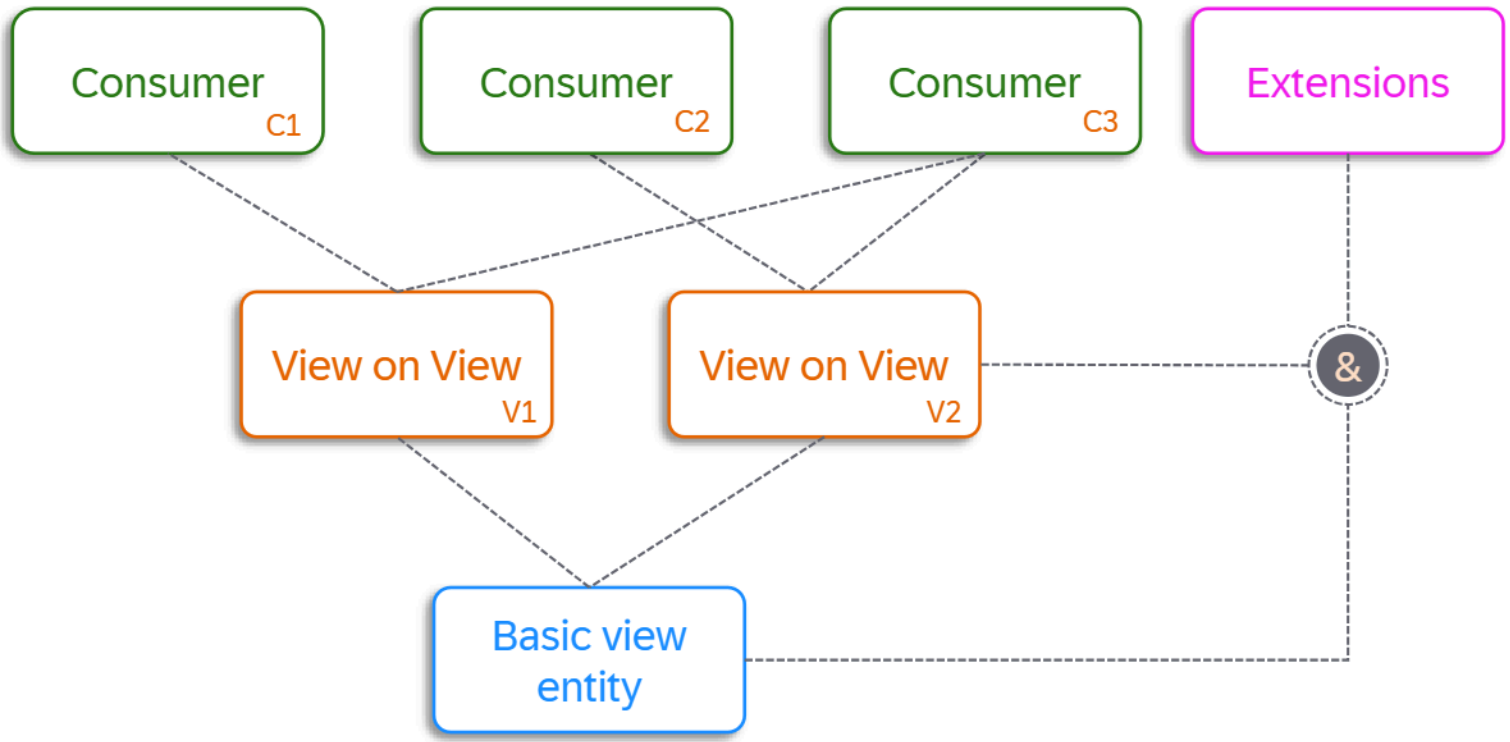
Reuse of generated joins when semantically identical

Can be queried in CDS views and ABAP SQL

- Navigate thru entity using path expressions



Nested CDS views and extensibility



Hierarchical view-on-view concept

Optimized result sets with minimum data transfer

Modification-free append of additional columns, arithmetic expressions or literals to result set

CDS hierarchies – Build hierarchies on SAP HANA

CDS Hierarchy defines a CDS entity as a hierarchy in the CDS DDL

A CDS hierarchy has a tabular results set whose rows construct parent-child relationships from hierarchical data sets

A CDS hierarchy can be accessed as the data source of an ABAP SQL query, and it is handled like a hierarchy in which additional hierarchy columns can be selected

```
define view CDS_SOURCE_VIEW
  as select from hierarchy_table_sp
  association [1..*] to CDS_SOURCE_VIEW as _Parent
//child to parent association (Mandatory)
  on  $projection.manager = _Parent.employee
 {
   key employee,
   key  manager,


 _Parent // Make association public


}
```

CDS source view

Self association

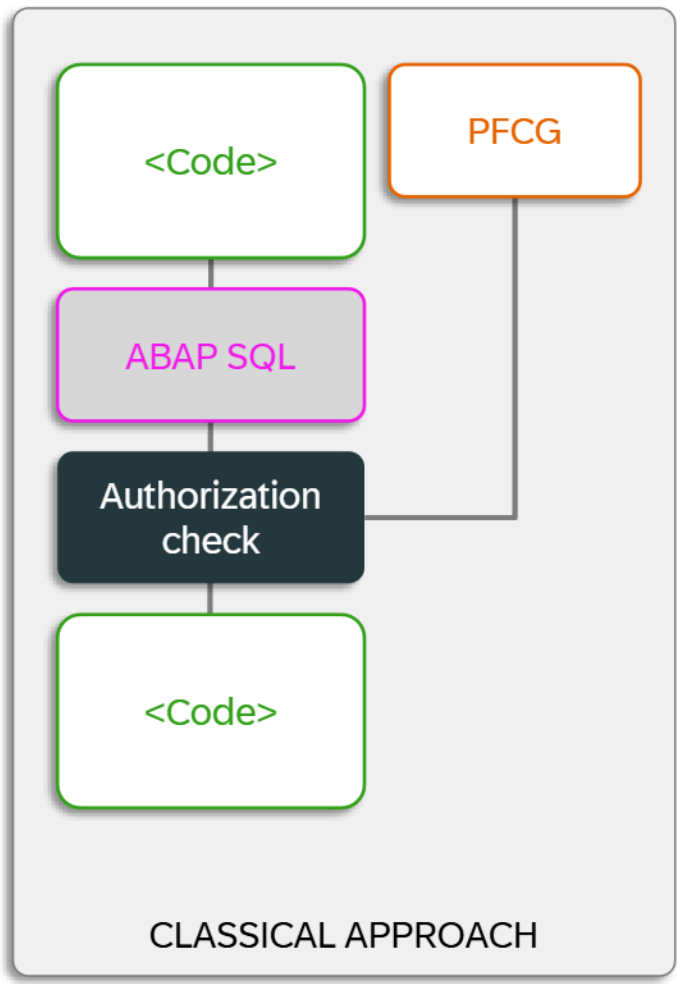
```
define hierarchy CDS_HIERARCHY
  as parent child hierarchy(
    source CDS_SOURCE_VIEW
   child to parent association _Parent
    start where
      SOURCE_CDS_VIEW.cost_center = 100
    siblings order by
      SOURCE_CDS_VIEW.employee
    multiple parents not allowed
  )
 {
   key employee,
   key manager,
  _Parent
 }
```

ABAP CDS hierarchy

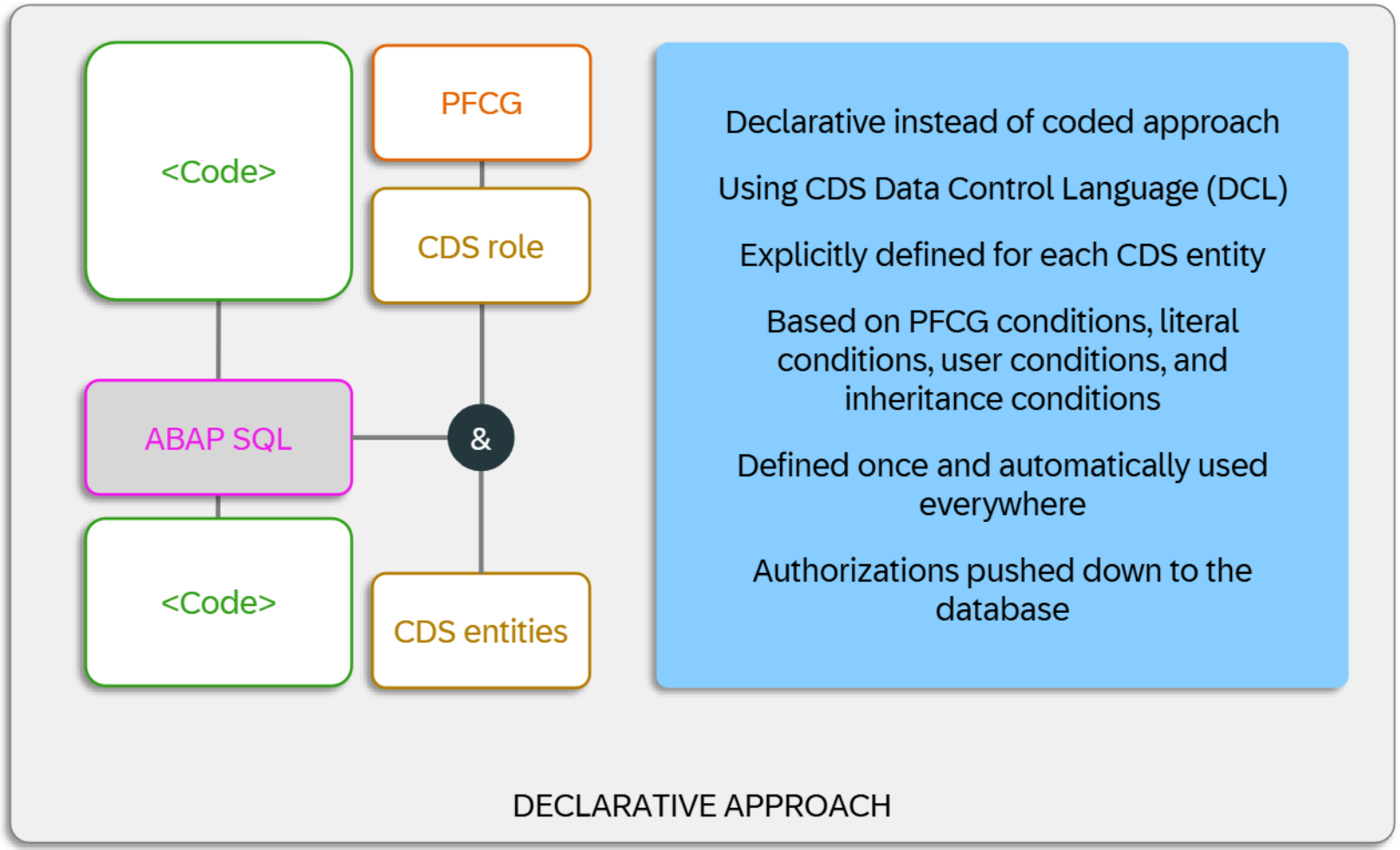
CDS source view

Filter results

Declarative access control with CDS roles







CDS development in Eclipse

01

Eclipse-based CDS Tooling

Source-based editor

Part of ABAP Development tools

02

Development support

Syntax check

Code completion

Syntax highlighting, pretty printing

Element info and navigation

Quick fixes

03

Supportability

Enhanced data preview

Dependency analyser

Activation graph / Dictionary log

Active annotations view

Unit testing

SAP HANA Breakouts

PUBLIC

Why code breakouts for SAP HANA make sense

UNLEASH THE FULL POTENTIAL OF YOUR SAP HANA DATABASE

Some scenarios require
selective measures

Highest performance requirements
e.g. with complex calculations

Use of database/analytical engine
specific functions required

ABAP SQL and plain CDS views are not sufficient to solve problem efficiently

ABAP managed database procedures (AMDP) for SAP HANA

UTILIZE NATIVE SAP HANA ENTITIES

Stored procedures, database functions and scalar functions supported

Expression of complex logic

Parameterized requests and multiple result sets

INTEGRATED IN THE ABAP INFRASTRUCTURE

Development, runtime error analysis, enhancement, transport

SAP HANA SQLScript coding embedded in ABAP classes

Seamless integration into CDS via CDS table functions

EASY ACCESS TO SAP HANA ADVANCED ENGINES / LIBRARIES

Like predictive analysis, financials, text mining, calculation engine

CDS table functions – Seamless AMDP integration into CDS

CDS table function definition

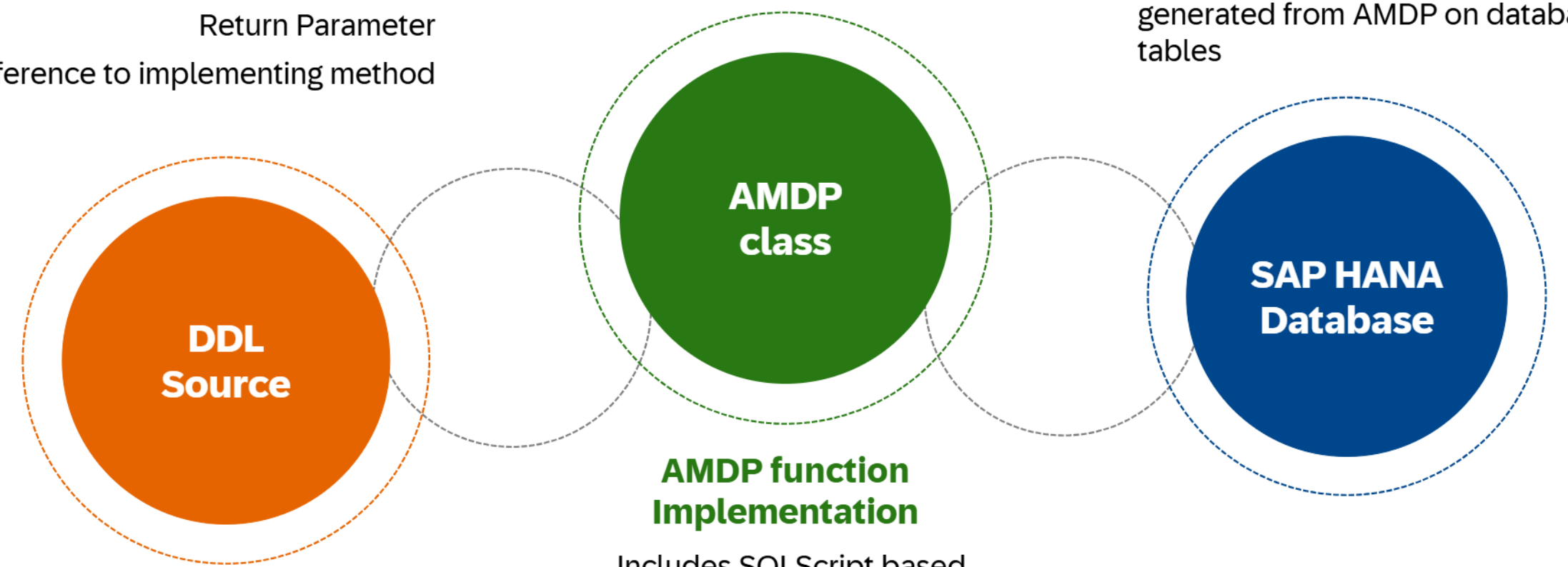
Parameter list

Return Parameter

Reference to implementing method

Runtime for table function

Runs stored SQLScript procedure generated from AMDP on database tables



AMDP function Implementation

Includes SQLScript based database function body

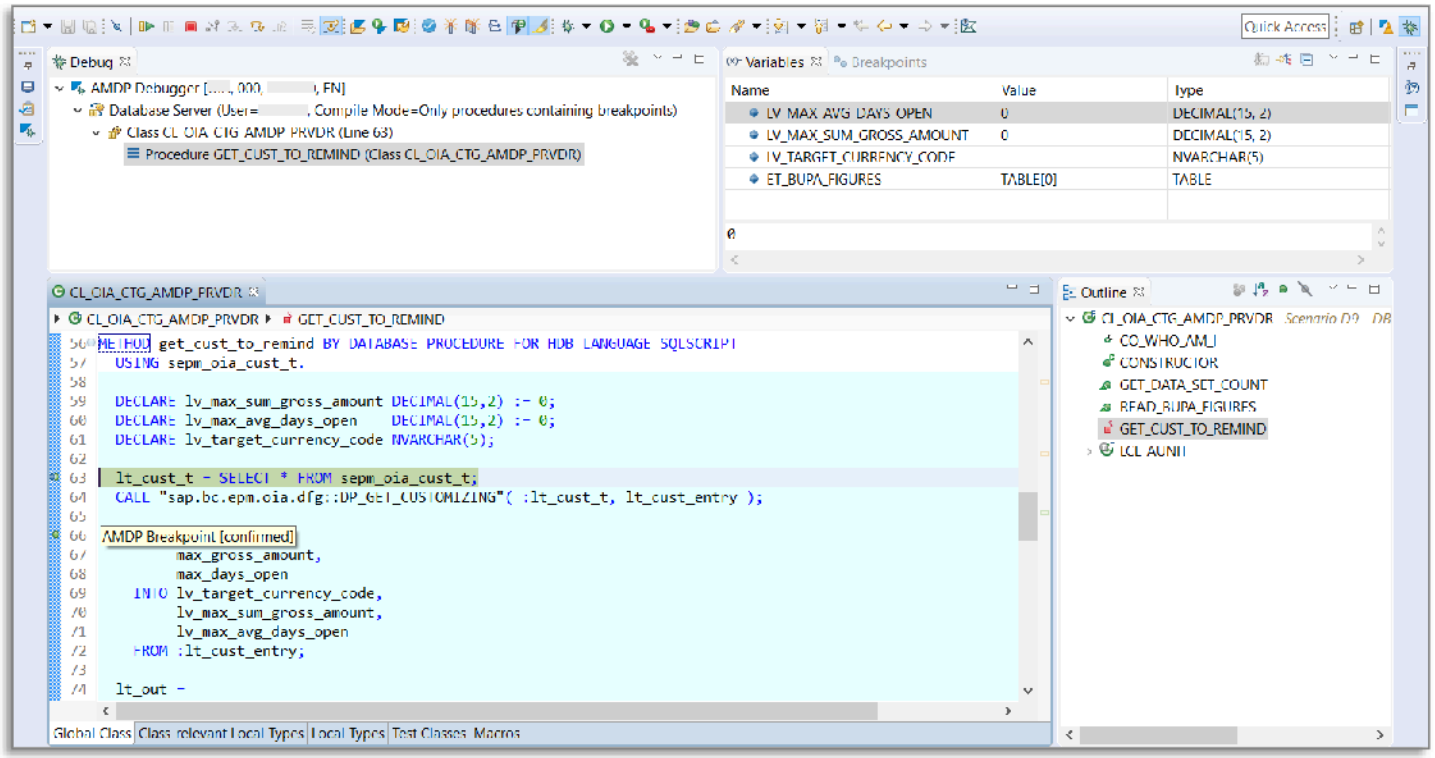
AMDP development in Eclipse

DEVELOPMENT

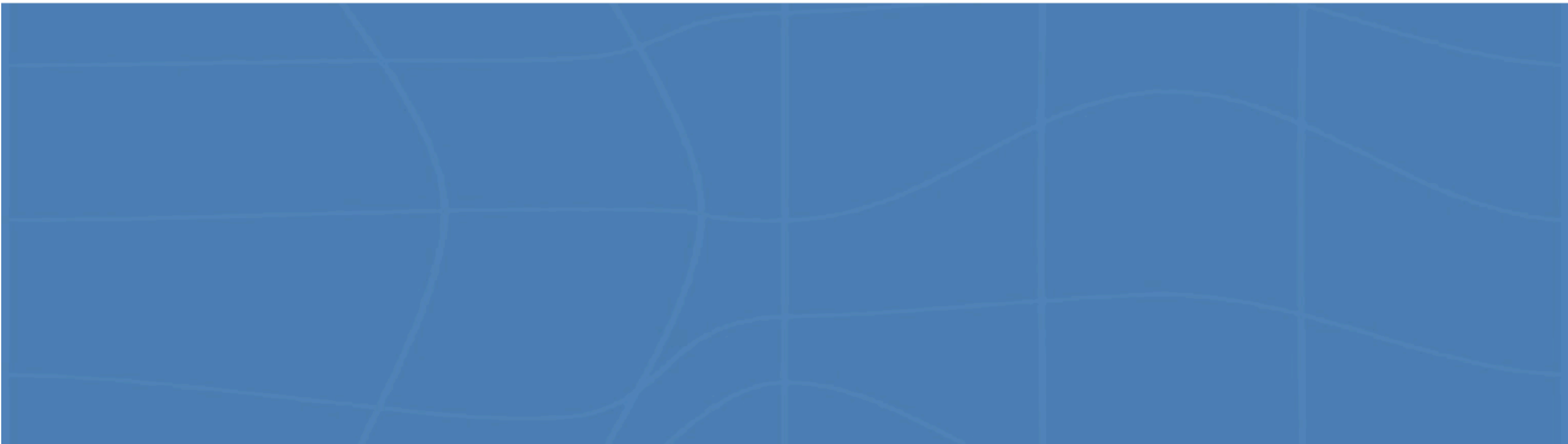
Eclipse-based source code editor
SQLScript syntax highlighting
Static syntax check
Error handling via class-based exceptions
Modification-free extension via AMDP BAdIs

SUPPORTABILITY

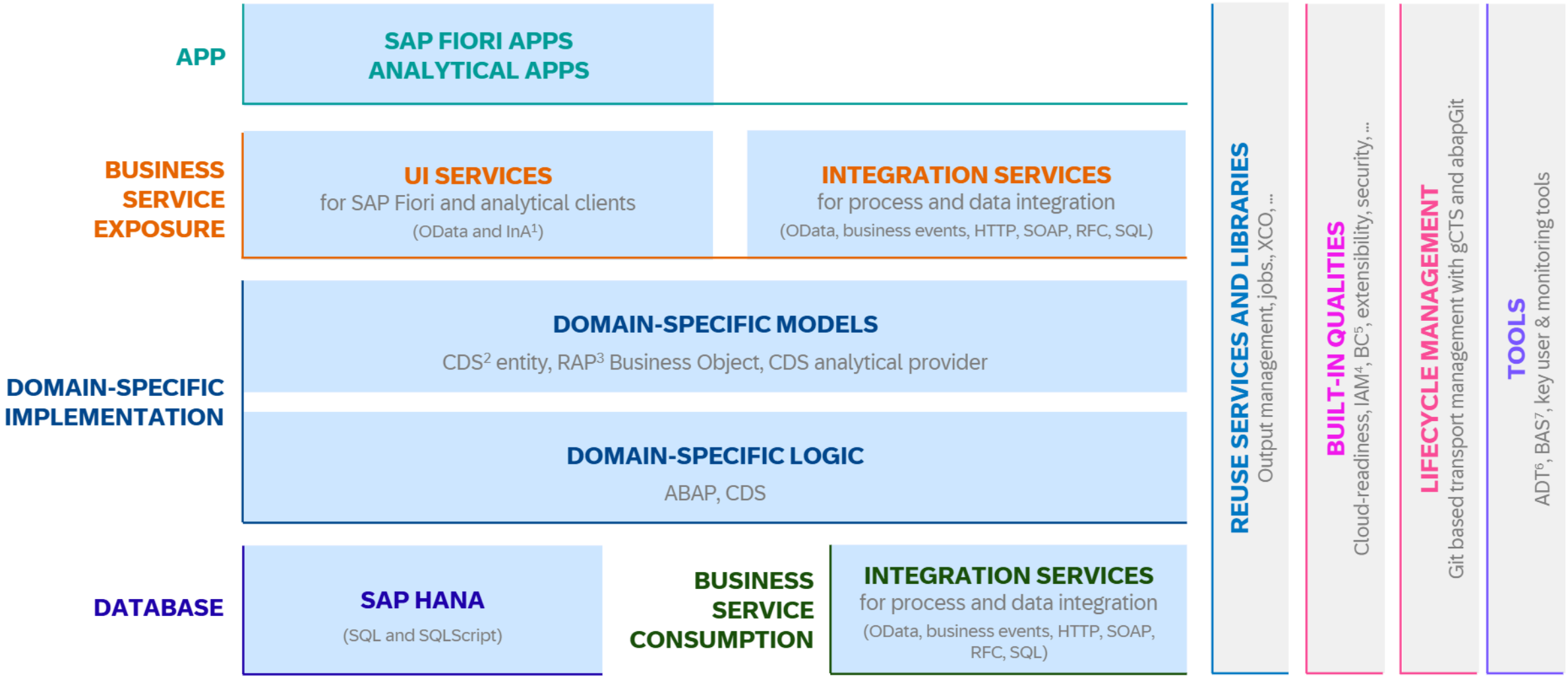
Autonomous AMDP debugger in ADT
Detailed runtime errors integrated in ST22 dumps



CORE DATA SERVICES IN ABAP APPLICATION PROGRAMMING



ABAP Cloud big picture



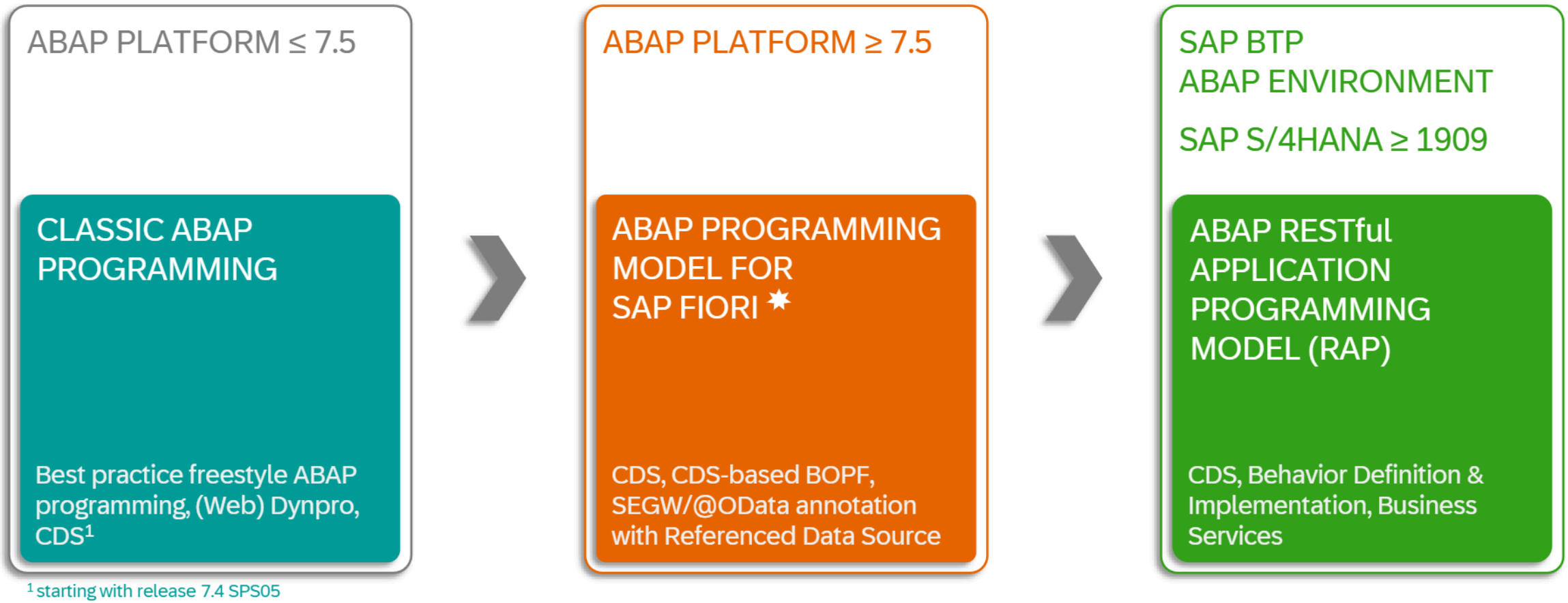
[1] Information access
[2] Core Data Services
[3] ABAP RESTful application programming model
[4] Identity & Access Management
[5] Business Configuration
[6] ABAP Development Tools
[7] Business Application Studio

ABAP RESTful APPLICATION PROGRAMMING MODEL (RAP)

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Evolution of the ABAP programming model

CDS as vital part of SAP S/4HANA and SAP BTP ABAP Environment

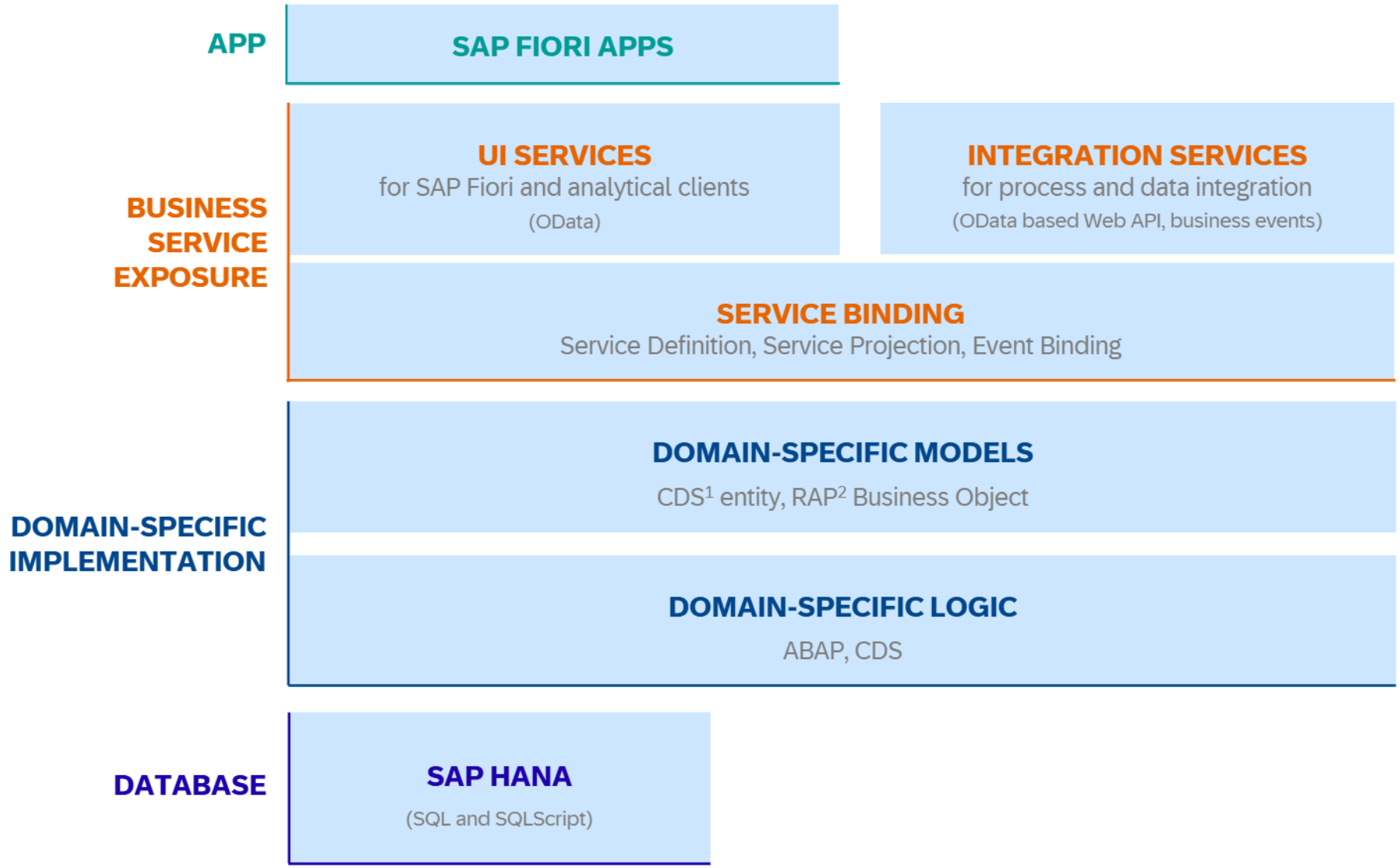


[1] starting with release 7.4 SPS05

✸ Safe investments!

ABAP RESTful application programming model – Big picture

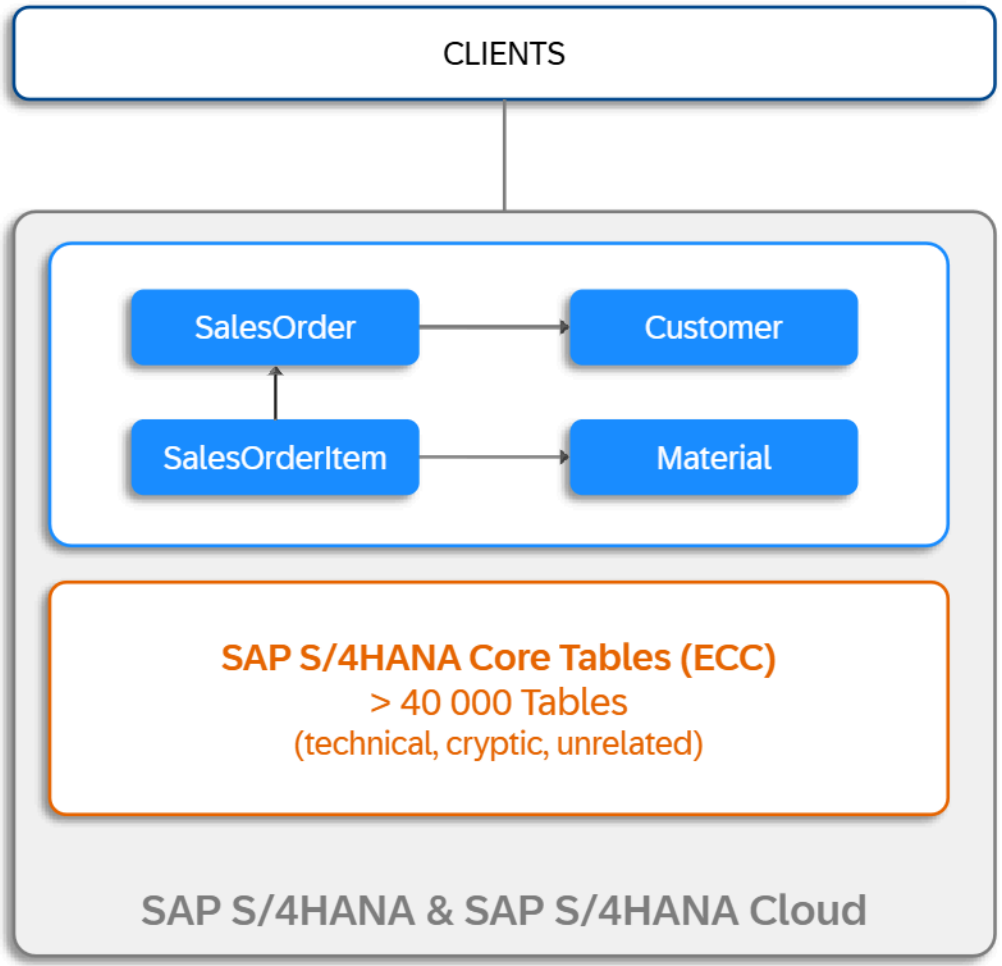
At the heart of ABAP Cloud for transactional scenarios



[1] Core Data Services

[2] ABAP RESTful application programming model

Virtual Data Model (VDM) for application data in SAP S/4HANA



VDM is a set of views on SAP S/4HANA application data

- Business-oriented
- Understandable & semantically rich
- Reusable & stable
- Executed on SAP HANA
- Implemented as ABAP CDS views

VDM is the data model and source for all types of apps

- Transactional
- Analytical
- External interfaces

Customers and partners can develop on released VDM views

CDS in SAP S/4HANA – Recommendations

- When implementing your own CDS views, follow the SAP S/4HANA's Virtual Data Model (VDM) implementation pattern for best reuse and interoperability
- Clarify your data model before starting your implementation
- Start with the implementation of basic interface views
  Adapt names and add semantical information already on this level

ANALYTICS

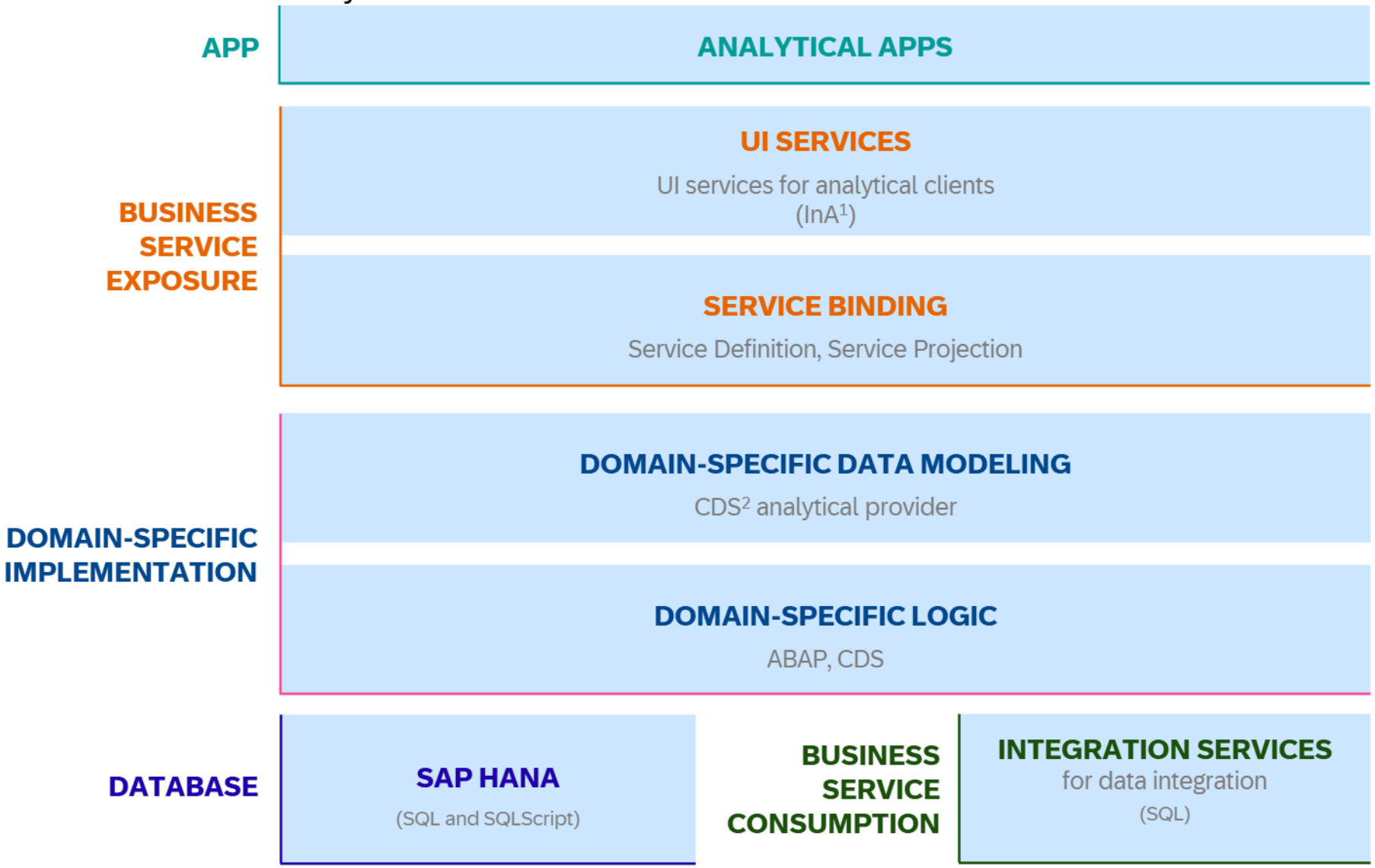
PUBLIC

Motivation for embedded analytics

	Transactional	Simple Analytics	Multi Dimensional	Embedded Analytics
Intention	Daily business tasks	Fast and easy insights	Interactive analysis	Daily business tasks Fast insights & access to interactive analysis
Interactivity	Linear	Static	Highly dynamic	On demand
Complexity	Low Single data sets	Low-medium Aggregating single sets	High Lots of data sets	Mixed
Aggregations	Very few	Yes	Dynamic: different types on different levels	All
Requests	Simple requests Few data	One request Complexity varies	Each interaction may trigger a complex request	Both
Request Types	CRUD	Read	Read	Both
Data Sources	Live data	Live data May be precalculated	Replicated data & live data	Live data

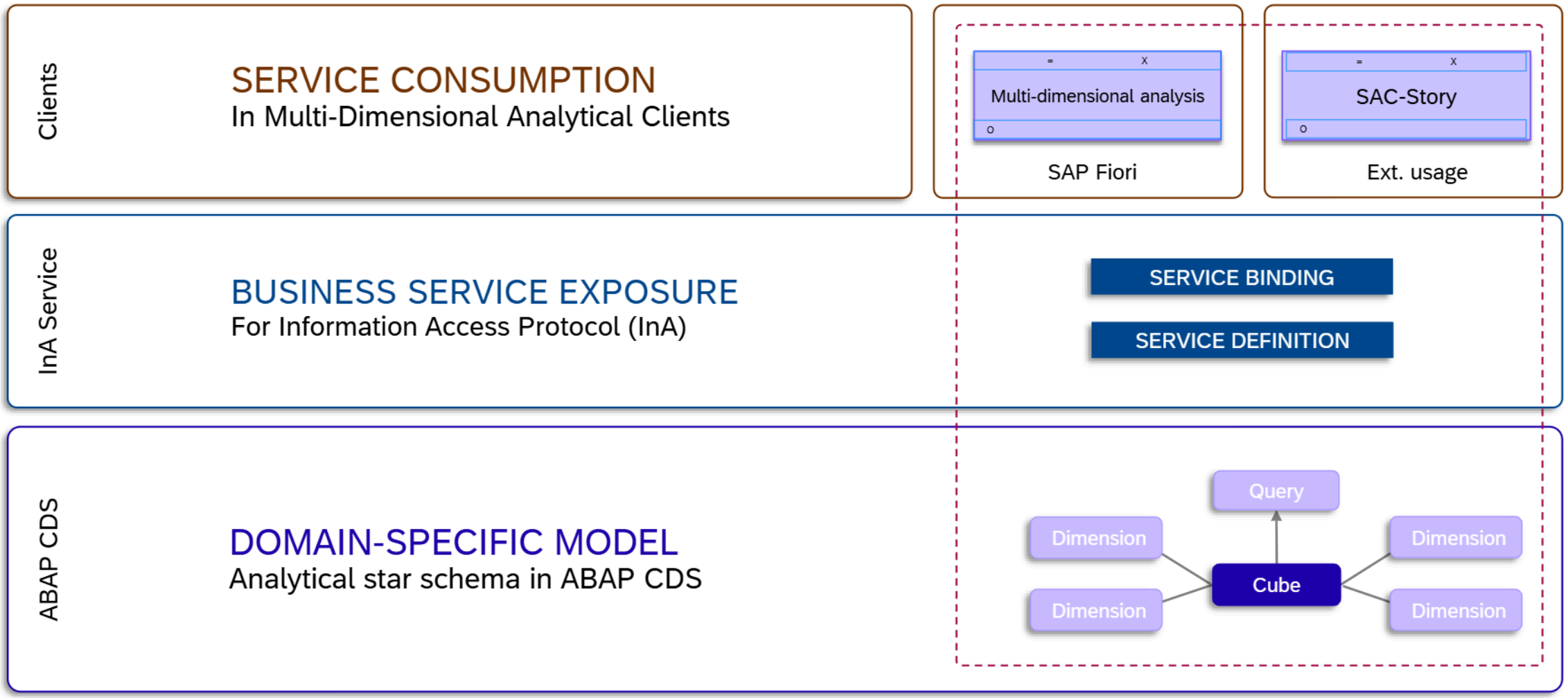
ABAP analytics – Big picture

At the heart of ABAP Cloud for analytical scenarios



[1] Information access
[2] Core Data Services

Standard development flow



SUMMARY

Key takeaways

ABAP Core Data Services (CDS)

- Code pushdown is the ABAP programming paradigm on SAP HANA
- CDS offers advanced capabilities to fully leverage SAP HANA's power in ABAP applications
- CDS is the next generation data modelling infrastructure on the ABAP platform
- CDS is the cornerstone of the ABAP RESTful Application Programming Model (RAP)
- CDS plays a vital role in SAP BTP ABAP environment and SAP S/4HANA on-premise as well as in the cloud
- Further enhancements and optimizations will be delivered with future ABAP platform releases

More information

SAP Community

- ABAP Development Community
- Landing pages: ABAP Platform 2021 | ABAP Platform 2020 | ABAP Platform 1909 | ABAP Platform 1809 | AS ABAP 7.52 | AS ABAP 7.51 | AS ABAP 7.50
- Getting started with ABAP development for SAP HANA
- Getting started with ABAP Core Data Services (CDS)
- Getting started with the ABAP RESTful Application Programming Model
- Developer center: https://www.sap.com/developer/topics/abap-platform.html

SAP Help Portal

- ABAP Platform 2021 | ABAP CDS Developer Guide | ABAP Keyword Documentation

SAP training and certification opportunities

- www.sap.com/education

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